CHARACTERIZATIONS OF BEING ZARISKI LOCALLY QUASI-SEPARATED THROUGH COVERS BY AFFINE SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the refinement of a cover into one with affine pieces over affine opens of the base, the deduction of quasi-compactness of the diagonal from the criterion on a surjective etale cover, and the reverse implication using the existence lemma and the quasi-compactness of the fibre products over the pieces.

Editorial note. The label lemma-quasi-separated-quasi-compact-pieces names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source spaces-properties.tex –label lemma-quasi-separated-quasi-compact-pieces –start 1064 –end 1149 –local-id 28043 returned RESULT=OK (L28043 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-spaces-properties/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Openness of the image of an etale morphism of algebraic spaces. Same chapter, the lemma that the image of an etale morphism of algebraic spaces is open. Supplies the open subspace X_ij corresponding to the image of the map from a piece of the affine cover.
- Quasi-compactness of a representable morphism tested on a surjective etale cover. Spaces, the lemma reducing a property of a representable morphism to a surjective etale cover by a scheme. Supplies the criteria by which quasi-compactness of the diagonal and of the morphisms U_i -> X_i is concluded.
- Existence of a Zariski cover by pieces mapping into affine opens with affine etale covers. Same chapter, the lemma producing a Zariski open covering by pieces mapping into affine opens of the base and admitting surjective etale maps from affine schemes. Supplies the covering used in the reduction from the Zariski local case.

Source reference. Original reference 06NF: sources/stacks-spaces-properties/pinned-original-source.tex, source lines 753–753. Prerequisite pointer invoked at proof

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

line(s) 1093; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03FY: sources/stacks-spaces-properties/pinned-original-source.tex, source lines 943–943. Prerequisite pointer invoked at proof line(s) 1125; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 01JO: sources/stacks-chow/tags.txt, source lines 2021–2021. Prerequisite pointer invoked at proof line(s) 1115; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02WK: sources/stacks-chow/tags.txt, source lines 3807–3807. Prerequisite pointer invoked at proof line(s) 1090; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02WM: sources/stacks-chow/tags.txt, source lines 3809–3809. Prerequisite pointer invoked at proof line(s) 1113; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02WZ: sources/stacks-chow/tags.txt, source lines 3822–3822. Prerequisite pointer invoked at proof line(s) 1119; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. The following are equivalent*

(1) *X is Zariski locally quasi-separated over S,*
(2) *X is Zariski locally quasi-separated,*
(3) *there exists a Zariski open covering $X = \bigcup X_i$ such that for each i there exists an affine scheme U_i and a quasi-compact surjective étale morphism $U_i \to X_i$, and*
(4) *there exists a Zariski open covering $X = \bigcup X_i$ such that for each i there exists an affine scheme U_i which maps into an affine open of S and a quasi-compact surjective étale morphism $U_i \to X_i$.*

Proof. Assume $U_i \to X_i \subset X$ are as in (3). To prove (4) choose for each i a finite affine open covering $U_i = U_{i1} \cup \ldots \cup U_{in_i}$ such that each U_{ij} maps into an affine open of S. The compositions $U_{ij} \to U_i \to X_i$ are étale and quasi-compact (see Spaces, Lemma 02WK). Let $X_{ij} \subset X_i$ be the open subspace corresponding to the image of $|U_{ij}| \to |X_i|$, see Lemma 06NF. Note that $U_{ij} \to X_{ij}$ is quasi-compact as $X_{ij} \subset X_i$ is a monomorphism and as $U_{ij} \to X$ is quasi-compact. Then $X = \bigcup X_{ij}$ is a covering as in (4). The implication (4) $\Rightarrow$ (3) is immediate.

Assume (4). To show that X is Zariski locally quasi-separated over S it suffices to show that X_i is quasi-separated over S. Hence we may assume there exists an affine scheme U mapping into an affine open of S and a quasi-compact surjective

étale morphism $U \to X$. Consider the fibre product square

$$\begin{array}{ccc} U \times_X U & \longrightarrow & U \times_S U \\ \downarrow & & \downarrow \\ X & \xrightarrow{\Delta_{X/S}} & X \times_S X \end{array}$$

The right vertical arrow is surjective étale (see Spaces, Lemma 02WM) and $U \times_S U$ is affine (as U maps into an affine open of S, see Schemes, Section 01JO), and $U \times_X U$ is quasi-compact because the projection $U \times_X U \to U$ is quasi-compact as a base change of $U \to X$. It follows from Spaces, Lemma 02WZ that $\Delta_{X/S}$ is quasi-compact as desired.

Assume (1). To prove (3) there is an immediate reduction to the case where X is quasi-separated over S. By Lemma 03FY we can find a Zariski open covering $X = \bigcup X_i$ such that each X_i maps into an affine open of S, and such that there exist affine schemes U_i and surjective étale morphisms $U_i \to X_i$. Since $U_i \to S$ maps into an affine open of S we see that $U_i \times_S U_i$ is affine, see Schemes, Section 01JO. As X is quasi-separated over S, the morphisms

$$R_i = U_i \times_{X_i} U_i = U_i \times_X U_i \longrightarrow U_i \times_S U_i$$

as base changes of $\Delta_{X/S}$ are quasi-compact. Hence we conclude that R_i is a quasi-compact scheme. This in turn implies that each projection $R_i \to U_i$ is quasi-compact. Hence, applying Spaces, Lemma 02WZ to the covering $U_i \to X_i$ and the morphism $U_i \to X_i$ we conclude that the morphisms $U_i \to X_i$ are quasi-compact as desired.

At this point we see that (1), (3), and (4) are equivalent. Since (3) does not refer to the base scheme we conclude that these are also equivalent with (2). □